

01 Aplicaciones Segunda Ley de Newton

$$-IAB \sin \theta = I_{\text{inercia}} \frac{d^2 \theta}{dt^2}$$

$$\theta \ll 1$$

$$\sin \theta \approx \theta$$

$$I_{\text{inercia}} \frac{d^2 \theta}{dt^2} = -IAB \theta$$

$$\frac{d^2 \theta}{dt^2} = -\frac{IAB}{I_{\text{inercia}}} \theta$$

$$\frac{d^2 \theta}{dt^2} = -\frac{I\pi R^2 B}{\frac{1}{2}mR^2} \theta$$

$$\frac{d^2 \theta}{dt^2} = -\frac{2I\pi B}{m} \theta$$

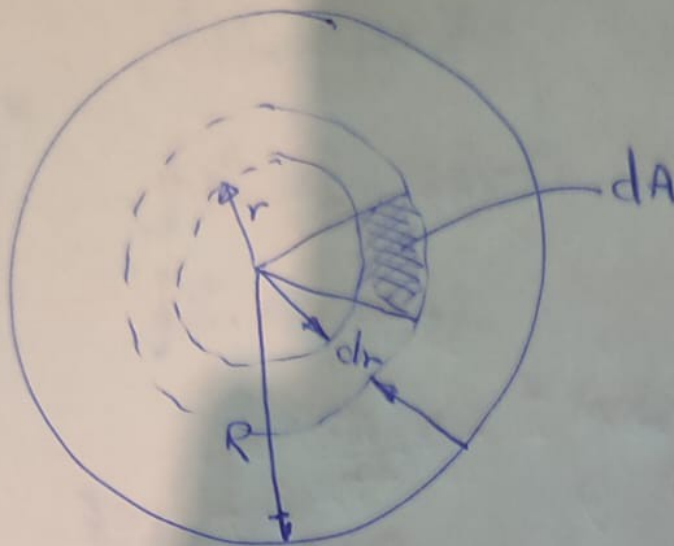
$$\frac{d^2 \theta}{dt^2} = -\omega^2 \theta$$

$$\omega = \sqrt{\frac{2\pi IB}{m}}$$

$$T = \frac{2\pi}{\omega}$$

$$T = \sqrt{\frac{2\pi m}{IB}}$$

Q2



$$dq = \sigma dA$$

$$dq = 2\pi \sigma r dr$$

$$dI = \frac{dq}{dt}$$

$$dI = \frac{2\pi \sigma r dr}{\frac{2\pi}{\omega}}$$

$$dI = \omega \sigma r dr$$

$$dB_x = \frac{\mu_0}{4\pi} \frac{2\pi r^2 dI}{(x^2 + r^2)^{3/2}}$$

$$dB_x = \frac{\mu_0 \omega \sigma r^3}{2(x^2 + r^2)^{3/2}} dr$$

$$B_x = \frac{\mu_0 \omega \sigma}{2} \int_0^R \frac{r^3}{(x^2 + r^2)^{3/2}} dr$$

$$B_x = \frac{\mu_0 \omega \sigma}{2} \left[\frac{R^2 + 2x^2}{\sqrt{R^2 + x^2}} - 2x \right]$$

Q3

$$d\phi_m = B dA \quad (1)$$

$$d\phi_m = B L dr$$

$$r < R$$

$$\oint \vec{B} \cdot d\vec{l} = 2\pi r B$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_c$$

$$B = \frac{\mu_0 I_c}{2\pi r} \quad (2)$$

$$\frac{I(r)}{I} = \frac{\pi r^2}{\pi R^2}$$

$$I(r) = I_c = I \frac{r^2}{R^2}$$

$$B = \frac{\mu_0 I}{2\pi r} \cdot \frac{r^2}{R^2}$$

$$B = \frac{\mu_0 I}{2\pi R^2} r$$

Ans [1]

$$d\phi_m = \frac{\mu_0 I}{2\pi R^2} r dr$$

$$\phi_m = \frac{\mu_0 I}{2\pi R^2} \int_0^R r dr$$

$$\phi_m = \frac{\mu_0 I}{4\pi}$$

$$\boxed{\frac{\phi_m}{L} = \frac{\mu_0 I}{4\pi}}$$

Q4

$$d\epsilon = B(dr) v$$

$$d\epsilon = B(dr)(rw)$$

$$d\epsilon = Bwrdr$$

$$\int_0^{\epsilon} d\epsilon = Bw \int_0^l r dr$$

$$\epsilon = \frac{1}{2} Bw l^2$$

$$|\epsilon| = \frac{d\phi_m}{dt}$$

$$dA = r\theta dr$$

$$A = \theta \int_0^l r dr$$

$$A = \frac{1}{2} \theta l^2$$

$$\phi_m = BA$$

$$\phi_m = \frac{1}{2} B l^2 \theta$$

$$|\epsilon| = \frac{d}{dt} \left[\frac{1}{2} B l^2 \theta \right]$$

$$|\epsilon| = \frac{1}{2} B l^2 \frac{d\theta}{dt}$$

$$\boxed{|\epsilon| = \frac{1}{2} B l^2 \omega}$$